

PercEF: progress update

Adaptive HNSW search when Ada-ef’s Gaussian model does not fit the data

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In one paragraph

HNSW answers every query with the same search breadth ef , although some queries need far more than others. Ada-ef (SIGMOD 2026) sets ef per query from a difficulty score whose thresholds come from a Gaussian model of a query’s similarities to the database. We measured how well that Gaussian fits on **41 public datasets** with a Kolmogorov–Smirnov (KS) statistic: it fits most modern text embeddings, but departs from the data on **9** (SIFT, GIST, Fashion-MNIST, DeepImage, Deep1B and other learned image/audio embeddings). There, Ada-ef’s score stops telling easy queries from hard ones. **PercEF** keeps Ada-ef’s mechanism but takes the thresholds from the data’s own empirical distance percentiles, with a 100-distance probe (Ada-ef: 1025) and an isotonic score-to- ef table; the search continues after the probe, nothing restarts. The same KS number, computed on the raw vectors in seconds, tells which of the two scores to use (right on 18 of 19 datasets).

How we compare

Every method is compared with a **hindsight-tuned fixed ef** : HNSW with one ef for all queries, chosen (interpolated on a dense grid) to reach exactly the method’s own mean recall. Against it we report work saved (distance computations), time saved (3 timing rounds), and the change in **p1**, the recall of the worst 1% of queries (Ada-ef’s own tail metric). 17 datasets are benchmarked end to end plus 3 cross-modal ones, each calibrated two ways: **P** (200 database vectors, Ada-ef’s protocol) and **R** (2000 held-out real queries). Target recall 0.95; $k = 100$ ($k = 1000$ on MS MARCO, Cohere, LAION). Same HNSW index ($M = 16$, $efC = 500$) for all methods; Ada-ef runs through its authors’ own code.

Headline counts (runs better than the tuned fixed ef)

Runs	Method	Less work	Faster	p1 $\geq$ fixed	Faster at equal p1
All 34 runs	PercEF	32/34	26/34	31/34	31/34
	Ada-ef	10/34	3/34	22/34	11/34
Non-Gaussian (16)	PercEF	14/16	8/16	13/16	13/16
	Ada-ef	2/16	0/16	6/16	1/16
Band (8)	PercEF	8/8	8/8	8/8	8/8
	Ada-ef	4/8	2/8	8/8	6/8
Near-Gaussian (10)	PercEF	10/10	10/10	10/10	10/10
	Ada-ef	4/10	1/10	8/10	4/10
Cross-modal (6)	PercEF	1/6	0/6	5/6	4/6
	Ada-ef	0/6	0/6	4/6	0/6

Less work / faster / p1: at the same mean recall. *Faster at equal p1*: against the fixed ef that gives the same worst-case recall. PercEF never needs more than 2.5% extra work in any of the 34 runs; Ada-ef up to 25%.

Every dataset (setting R)

Dataset	Ranking $ \rho $		Ada-ef vs tuned fixed ef				PercEF vs tuned fixed ef			
	KS	Ada / PercEF	recall	work	time	p1	recall	work	time	p1
<i>Non-Gaussian ($KS \geq 0.066$): where PercEF is meant to help</i>										
SIFT-1M	0.126	0.04 / 0.45	0.963	-0.2	-25.2	-0.0	0.964	+2.7	+0.4	+2.8
iNat-ResNet	0.115	0.21 / 0.46	0.969	-7.5	-101.3	+0.2	0.976	+4.3	+3.1	+3.0
GIST-960	0.091	0.08 / 0.23	0.952	-5.7	-10.8	-4.6	0.951	-0.8	-1.6	-1.5
Yambda	0.090	0.03 / 0.43	0.978	-5.7	-39.9	-0.2	0.983	+2.4	+2.7	+1.7
Landmark-DINO	0.076	0.34 / 0.55	0.955	-6.2	-36.5	-0.1	0.968	+4.0	+1.8	+3.9
Fashion-MNIST	0.073	0.17 / 0.29	0.994	-24.6	-60.4	-1.0	0.992	+6.2	+4.5	+0.5
Deep1B (2M)	0.067	0.41 / 0.67	0.961	+0.3	-23.1	+5.2	0.961	+3.7	-2.4	+8.1
DeepImage-96	0.066	0.38 / 0.62	0.966	+1.2	-11.5	+4.8	0.958	+0.5	-7.1	+8.8
<i>Crossover band ($0.044-0.066$)</i>										
Yahoo-MiniLM	0.054	0.50 / 0.68	0.965	+0.5	-20.5	+6.5	0.964	+3.4	+0.5	+4.7
Cohere-1024	0.049	0.79 / 0.52	0.967	+1.1	-9.2	+9.4	0.965	+4.7	+2.3	+1.7
COCO-I2I	0.046	0.35 / 0.36	0.985	-5.4	-37.6	+2.6	0.985	+3.7	+0.3	+1.4
MS MARCO-384	0.044	0.69 / 0.53	0.972	+5.2	+0.5	+6.8	0.972	+5.1	+3.8	+2.6
<i>Near-Gaussian ($KS < 0.044$): Ada-ef's home ground</i>										
DBpedia-1536	0.038	0.48 / 0.54	0.963	-0.5	-45.5	+3.3	0.967	+5.3	+2.0	+3.4
LAION-I2I	0.034	0.57 / 0.53	0.958	-7.3	-10.8	+0.0	0.956	+11.1	+6.0	+0.3
BIGANN (2M)	0.029	0.42 / 0.74	0.960	+0.2	-15.5	+1.9	0.964	+7.0	+1.4	+5.7
GloVe-100	0.018	0.80 / 0.61	0.953	+11.5	+0.0	+7.3	0.943	+5.9	+1.4	+2.9
MS Turing (2M)	0.010	0.48 / 0.40	0.952	+0.1	-1.2	-4.1	0.950	+1.9	+0.9	+1.0
<i>Cross-modal queries (neither method adapts)</i>										
Last.fm	0.216	—	0.974	-10.3	-86.4	+2.8	0.971	-0.2	-6.5	+0.5
ImageNet-ALIGN	0.058	—	0.951	-0.0	-19.0	+0.0	0.963	-1.0	-0.9	+0.5
COCO-T2I	0.056	—	0.964	-0.2	-16.0	+0.6	0.959	-0.2	-0.9	+1.3

recall: the method's mean recall (the fixed ef is tuned to it). *work*, *time*: % saved over the tuned fixed ef (positive = better). *p1*: change in worst-1% recall, in points. *Ranking*: rank correlation between each score and the true ef each calibration query needs (higher = better). Bold: PercEF better than the fixed ef . Setting P gives the same picture (counts above).

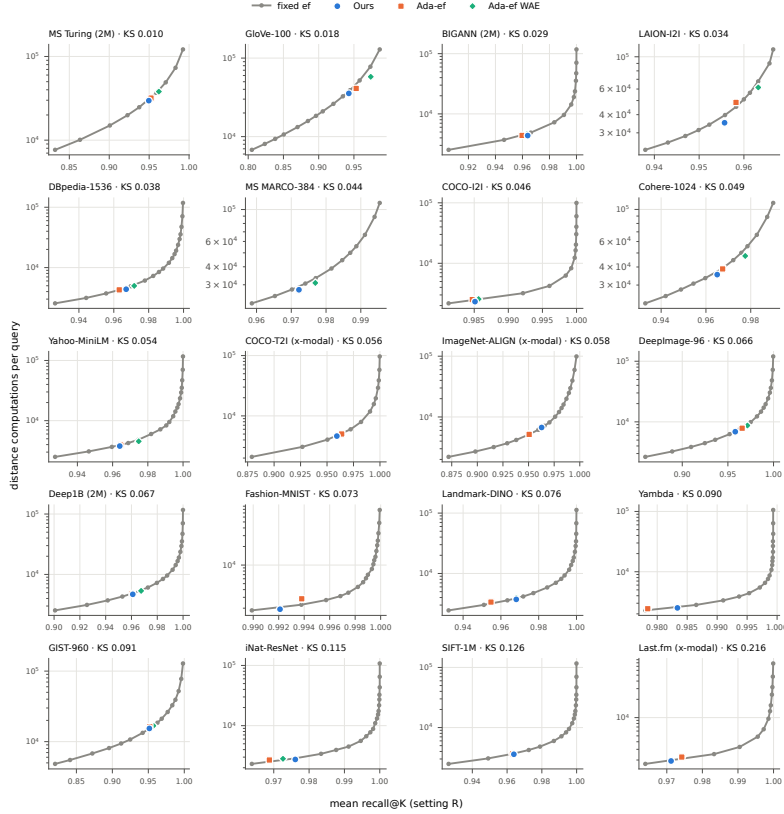
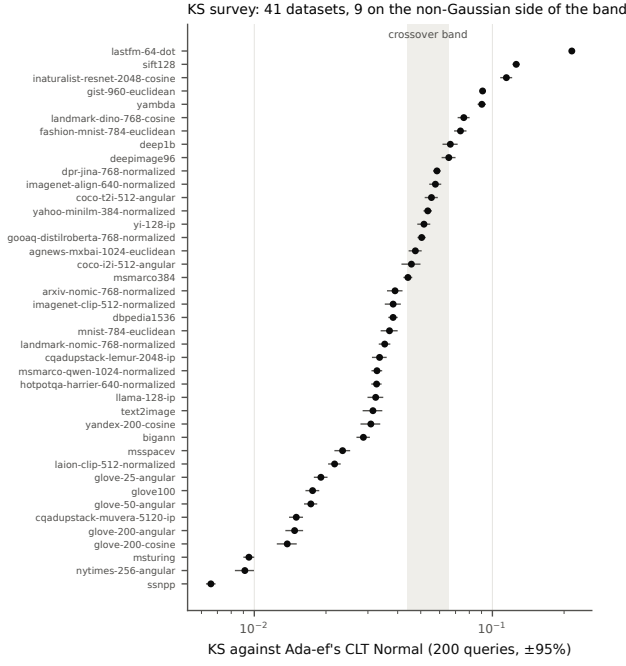
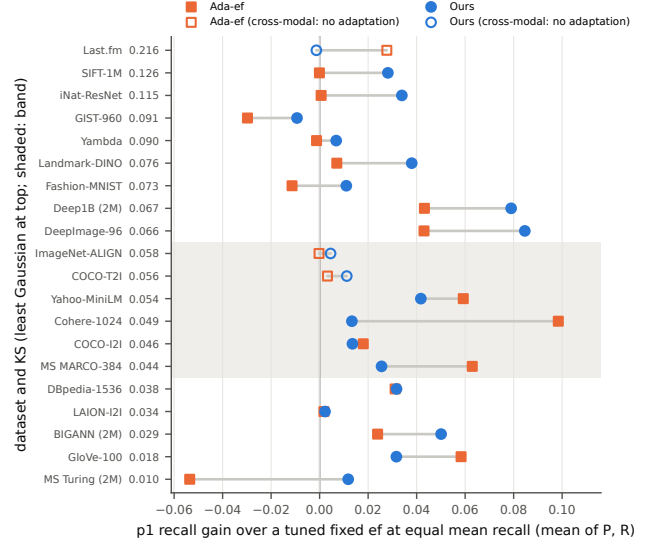


Figure 1: Distance computations per query vs mean recall (setting R). Grey: fixed ef ; points below the curve reach the same recall with less work.



(a) KS between each query's similarities and Ada-ef's Gaussian, 41 datasets (mean of 200 queries, 95% interval).



(b) p1 gain over the tuned fixed ef at the same mean recall (mean of P and R), ordered by KS. Above the shaded band PercEF's gain is larger; below it Ada-ef's usually is.

Other checks

- **Other targets** (0.90, 0.99) and $k = 10$ on 6 datasets: the comparisons hold at 0.90 and 0.99 (non-Gaussian: PercEF less work in 7/8 and 6/8 runs, faster at equal p1 in 8/8; Ada-ef 0/8 and 2/8). At $k = 10$ neither method saves work over the tuned fixed ef .
- **Meeting the target recall:** one offline factor on PercEF's table (fitted on calibration queries) makes it land within 0.01 of the target at least as often as Ada-ef.
- **Probe length ablation:** the gain comes from the thresholds, not the shorter probe.
- **DARTH** (learned early termination, SIGMOD 2026) on 4 datasets: best tail, but $1.9\text{--}2.3\times$ slower than a fixed ef ; PercEF within 3%.
- **Why time lags work on non-Gaussian data:** hard queries cost about 8% more time per distance even under a fixed ef , and PercEF moves work to them.

Limits

On near-Gaussian text embeddings Ada-ef gains more worst-case recall (PercEF is still cheaper and faster than the fixed ef there); at $k = 10$ and with cross-modal queries neither method adapts; one KS prediction (SIFT-1B) is wrong.

Status

The paper (8 pages, *Experiments & Analysis*) is registered at **SIGMOD 2027, research round 4**; the full paper is due **17 October 2026**. Everything is reproducible from the released result files (code: HNSWlib build with Ada-ef's own code, benchmark and analysis scripts).